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Academic Year: 2026-27	**BE Project Synopsis**	**Revision : 00 Dated : 20/11/2019**
Term – I	**Department: Artificial Intelligence and Data Science**	**Date of Preparation : 19/08/2026**

Project Title	NeuroShield: A Dual-Core Explainable AI Intrusion Detection System (IDS) for Multi-Stage Threat Prediction in Smart Manufacturing
Domain Name	Cybersecurity / Artificial Intelligence & Data Science (Explainable AI for Industrial IoT & IT-OT Security)
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ABSTRACT:

NeuroShield is an advanced Explainable AI-based Intrusion Detection System (IDS) for Smart Manufacturing environments. It combines a CNN-LSTM network engine for IIoT traffic analysis with a GNN-based engine for device and process behavior monitoring. Using SHAP, LIME, and GNNExplainer, it provides transparent explanations for every prediction, while a Trust Validator generates a reliable Trust Score. By correlating alerts with MITRE ATT&CK and predicting the next attack stage, NeuroShield enables proactive and trustworthy cyber threat detection.

Keywords: Smart Manufacturing, Industrial IoT (IIoT), IT-OT Security, Explainable AI, CNN-LSTM, Graph Neural Network, SHAP, LIME, GNNExplainer, Trust Score, Attack-Chain Prediction

MOTIVATION:

- Smart manufacturing connects IT with OT systems such as PLCs, SCADA, and IIoT sensors, creating security risks where compromised IT devices can directly impact production controllers.
- Distributed attacks like DDoS, probing, and botnets can disrupt IIoT networks, while abnormal API and process behavior can compromise individual devices without triggering traditional network-level alarms.
- Conventional IDS and XAI methods often lack transparency and explanation reliability, making it difficult for OT operators to trust and act on security predictions.

OBJECTIVES:

- To design a passive, real-time monitoring mechanism using Port Mirroring (SPAN) that analyzes network logs without causing operational disruption to critical OT assets like PLCs and robotic arms.
- To utilize unsupervised learning models like Graph Neural Networks to establish a baseline of normal factory behavior, enabling the instant detection of zero-day anomalies and IT-to-OT lateral movement.
- To integrate an Explainable AI (XAI) layer utilizing LIME, SHAP, or GNNExplainer to translate complex machine learning anomalies into plain-English, actionable alerts on a web dashboard.
- To prevent cyber-physical sabotage by continuously monitoring the exact network commands sent to heavy machinery, instantly flagging malicious instructions before they can trigger a physical disaster.
- To establish an external protective net for fragile, unpatchable legacy machinery by analyzing traffic at the network switch level, delivering robust AI security without requiring local software agents.

ARCHITECTURE DIAGRAM:

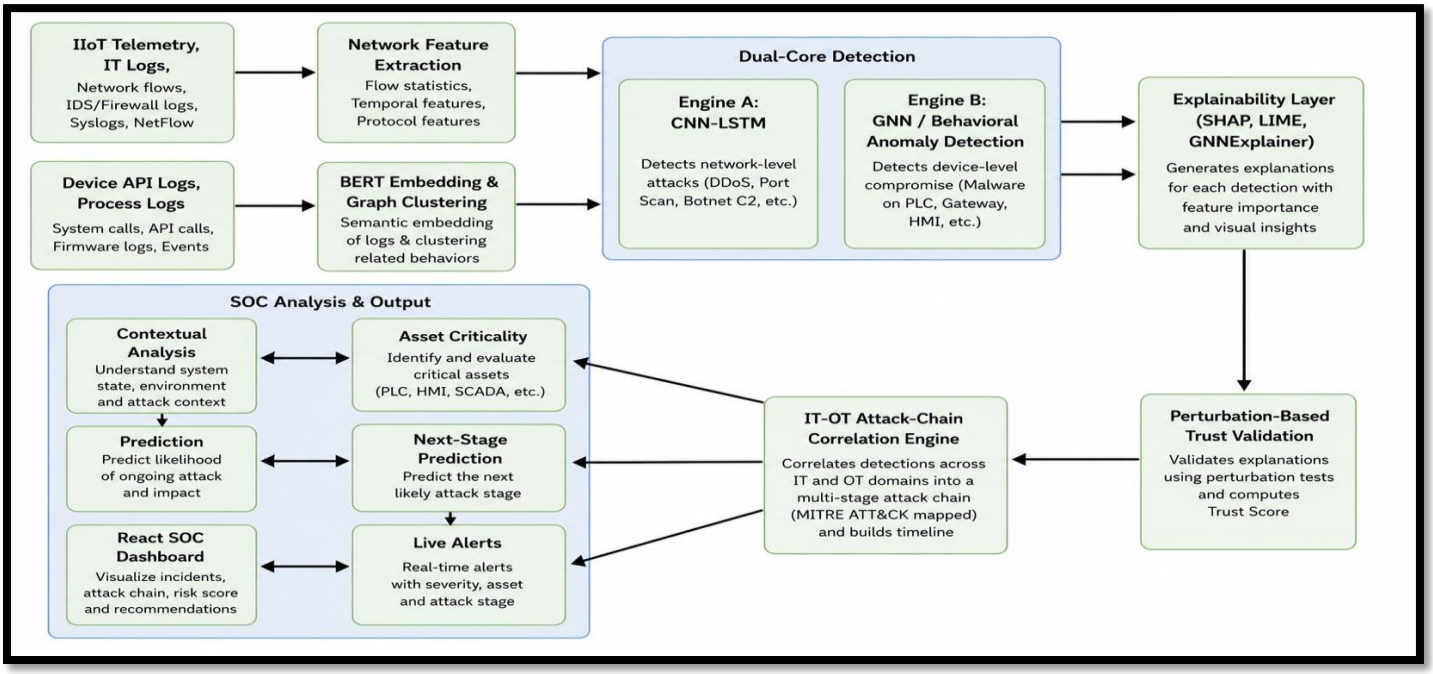


Fig.: System Architecture of "NeuroShield"

APPLICATION AREAS:

- Smart Factory Security: Protects PLCs, SCADA, and workstations from compromise and IT-to-OT lateral movement.
- Industrial MSSPs & SOCs: Provides transparent, multi-stage threat detection as a managed service for manufacturing clients.
- OT Security Vendors: Embeds dual-engine detection into edge gateways to intercept threats before production impact.
- Large-Scale IIoT: Extends attack correlation capabilities across multi-site manufacturing environments.
- Regulatory Compliance: Assists manufacturers in meeting auditable AI-security standards like the NIST framework.

CONCLUSION:

NeuroShield addresses the critical IT-OT security risks in Smart Manufacturing by combining a CNN-LSTM Network Attack Engine and a GNN-based Device Behaviour Engine to detect both distributed and individual-device cyberattacks. It replaces traditional black-box AI with transparent, human-readable predictions using SHAP, LIME, and GNNExplainer, further validated by a novel Trust Score. By correlating these cross-domain alerts into a single MITRE ATT&CK chain with a predicted next stage, NeuroShield transforms industrial cybersecurity from a reactive monitoring tool into a trustworthy, early-warning defense system

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